A crude gradient Descent Algorithm for Minimizing a Function

- Let $f: \mathbb{R}^n \to \mathbb{R}$ be a function we want to minimize.

- Start with any point x^0. Generate a sequence of points $x^1, x^2, \ldots$, by moving against the gradient for small step sizes.

- Say at iteration k, we are at point x^k. The next point is computed as

$$x^{k+1} = x^k - \alpha^k \cdot Df(x^k),$$

where α^k is a "small" step size

- How small is small? Do trial and error. Pick any α^k and check if

$$f(x^k - \alpha^k Df(x^k)) < f(x^k)$$

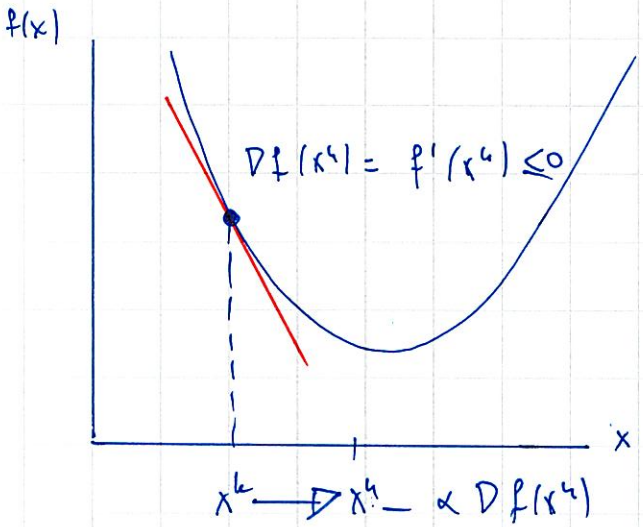
If so, take the step
If not, halve α^k and try again.

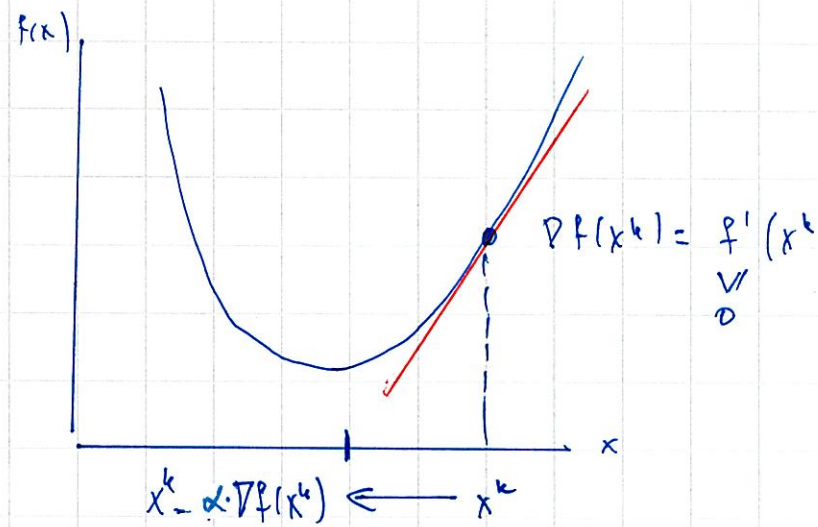
- Say we want to minimize

$$f(x) = 4 \cdot \ln\left(e^{2x_1} + e^{4x_2}\right) - 6x_1 - 4x_2$$

$$\nabla f(x) = \left(4 \frac{2 \cdot e^{2x_1}}{e^{2x_1} + e^{4x_2}} - 6 \,,\; 4 \frac{4 \cdot e^{4x_2}}{e^{2x_1} + e^{4x_2}} - 4\right)$$

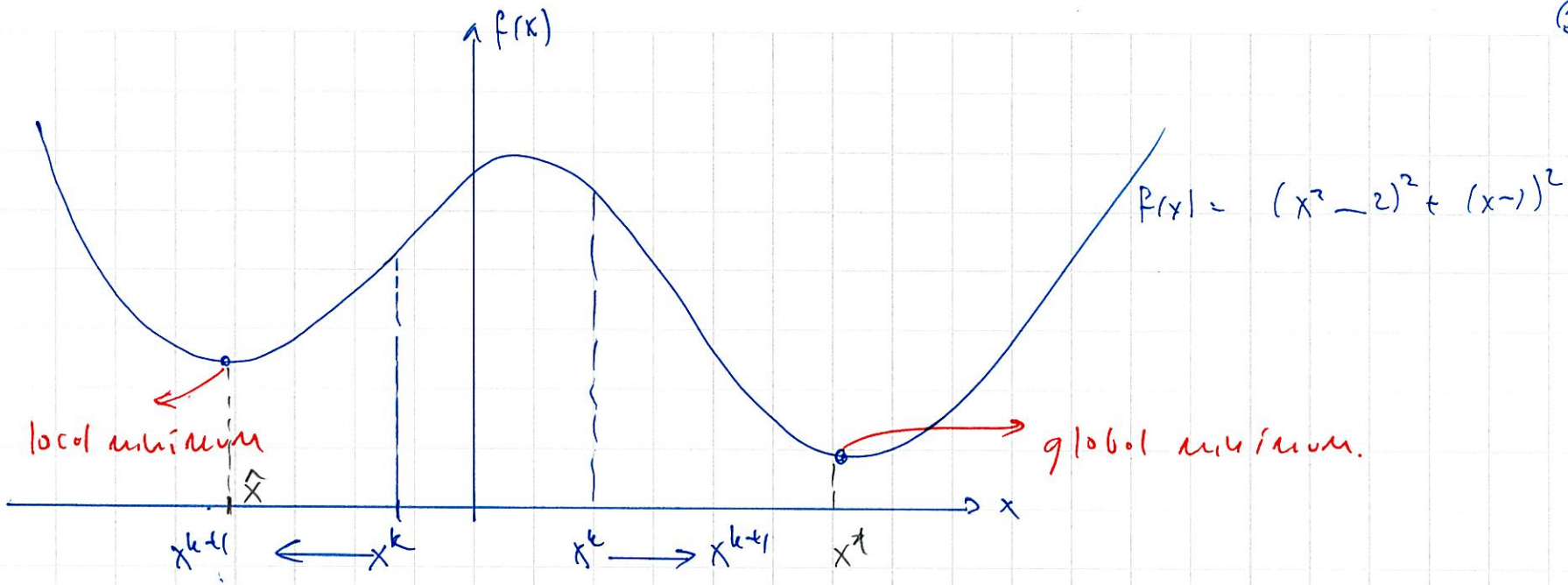
- What is the best we can expect from gradient descent?





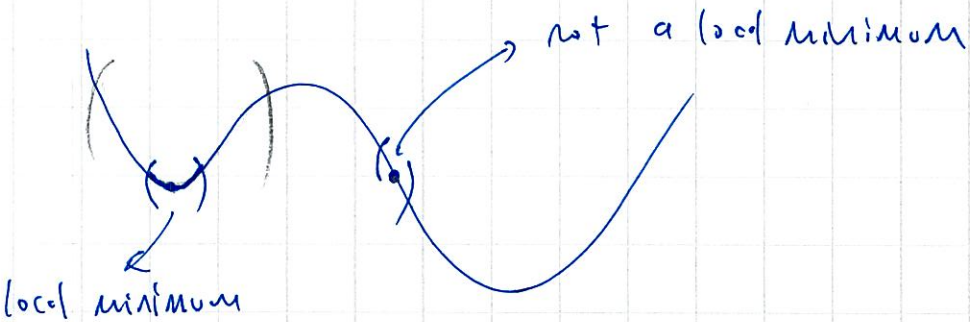
$$x^k - \alpha \nabla f(x^k) \geq x^k$$

$$x^k - \alpha \underset{\geq 0}{\nabla f(x^k)} \leq x^k$$



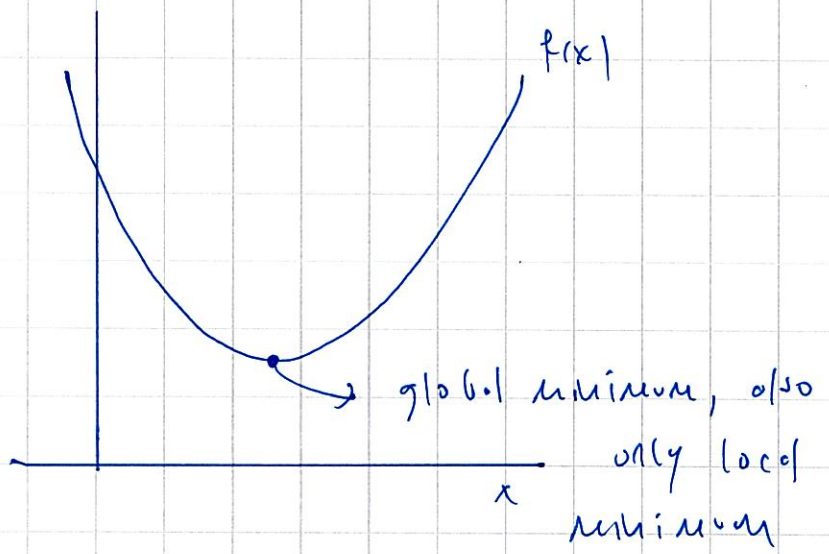
- global minimum x^* — x^* satisfies $f(x^*) \leq f(x)$ for any x

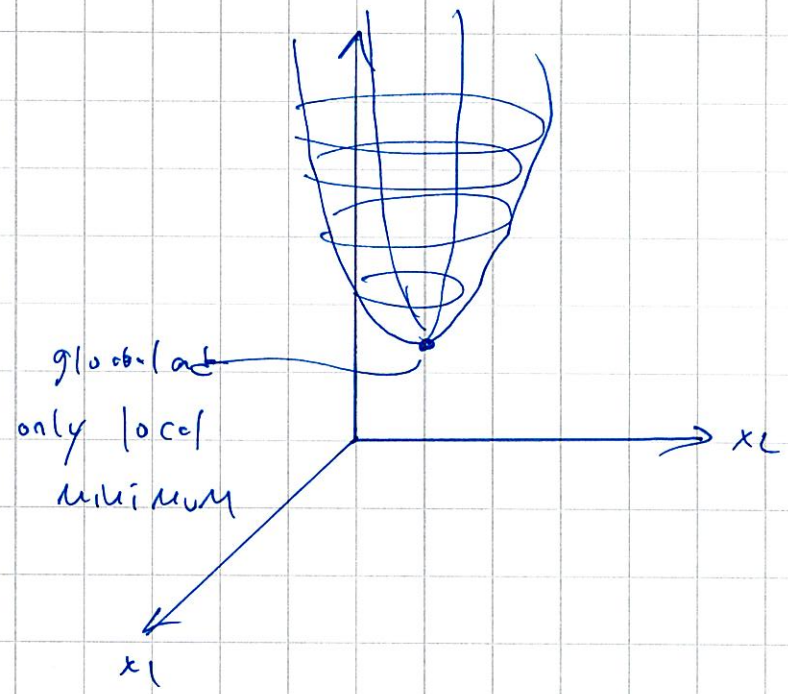
  local minimum $\hat{x}$ — $\hat{x}$ satisfies $f(\hat{x}) \leq f(x)$ for any x in a small neighborhood of $\hat{x}$

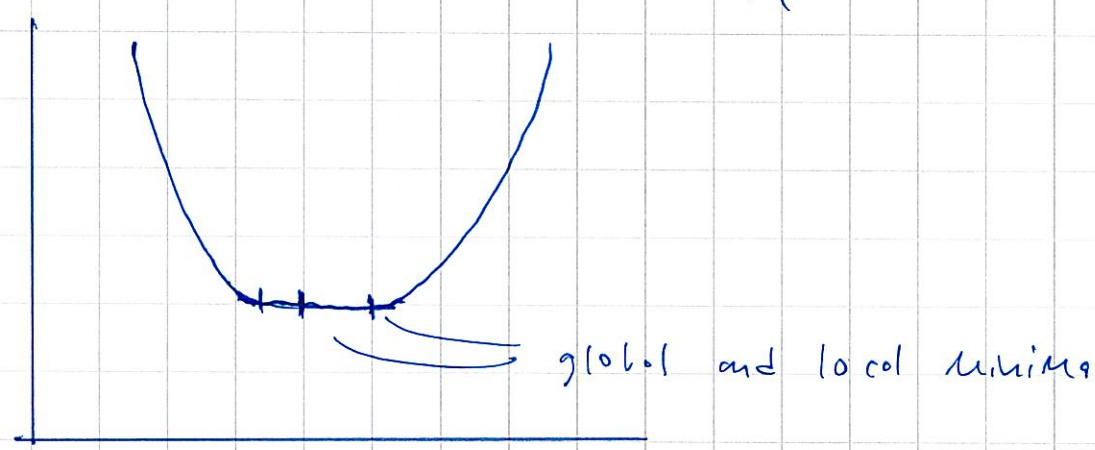


- Best we can expect from gradient descent is we get to a local minimum.

- If we are minimizing a convex function, then there is no distinction between global and local minimum. All local minima are also global minima







- If we apriori know that the objective function we are minimizing is convex, then we can be sure about the success of the gradient descent algorithm

- When the obj. function is not convex, a heuristic method to try to find a global minimum is to start gradient descent with different initial points

- Let's focus on the step size selection problem: Currently we are at point x^k. We computed $\nabla f(x^k)$.

- Choose the step size α such that the value of the function at the next point we reach is as small as possible. In other words, solve the optimization problem:

$$\min_{\alpha \geq 0} f\big(\underbrace{x^k}_{\text{known}} - \alpha \underbrace{\nabla f(x^k)}_{\text{known}}\big)$$

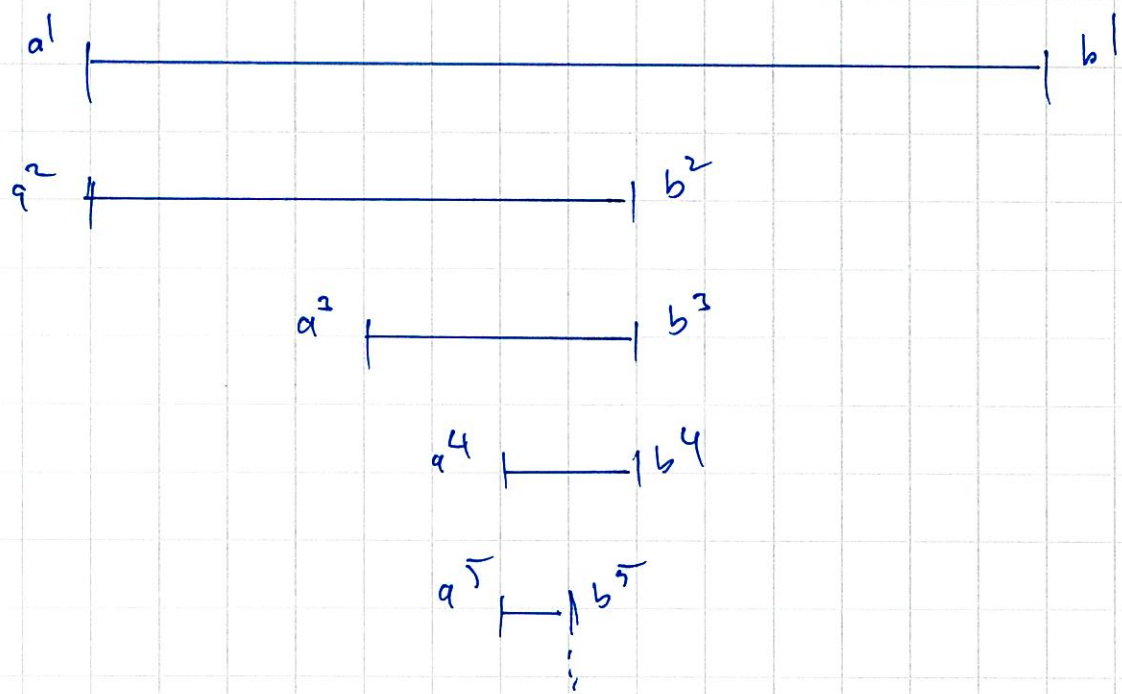
This optimization problem is a single-dimensional optimization problem over α

Single Dimensional Optimization

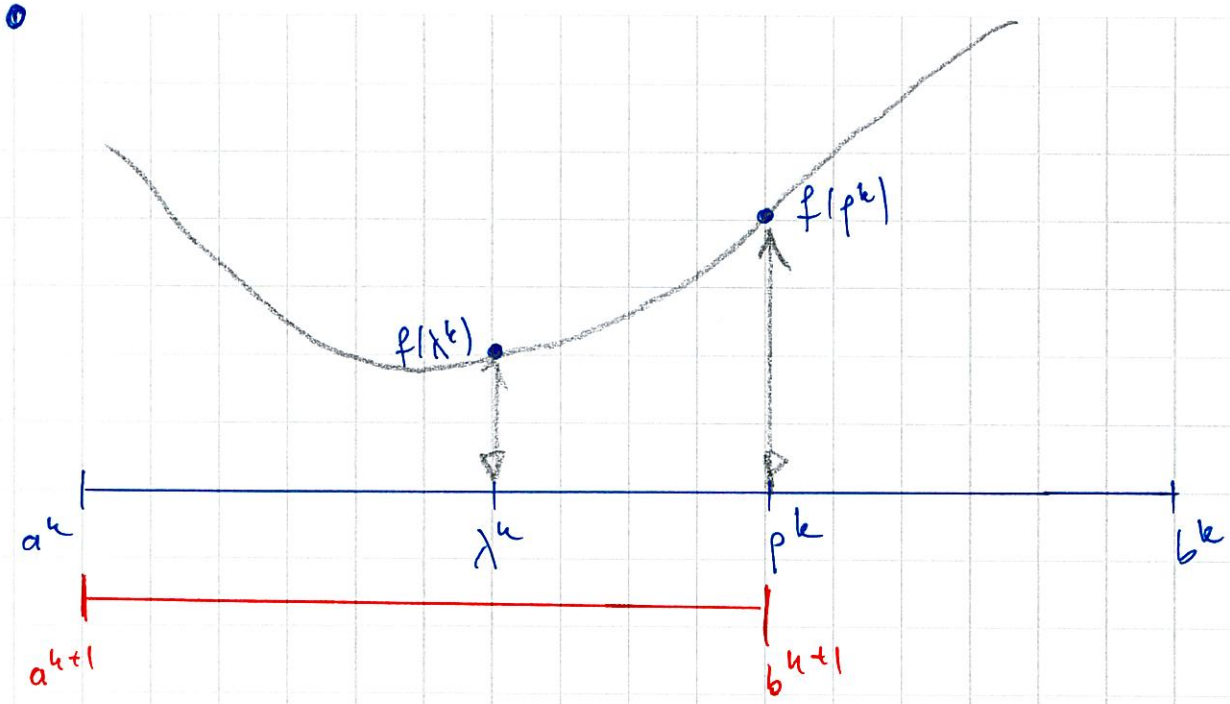
- We have a function $f: \mathbb{R} \to \mathbb{R}$ and we want to minimize f.

- We will construct iterative algorithms to minimize f.
  We will start with an interval of uncertainty $[a^1, b^1]$ over which we want to find the minimizer of f.

- Iteratively, we will reduce the width of the interval of uncertainty

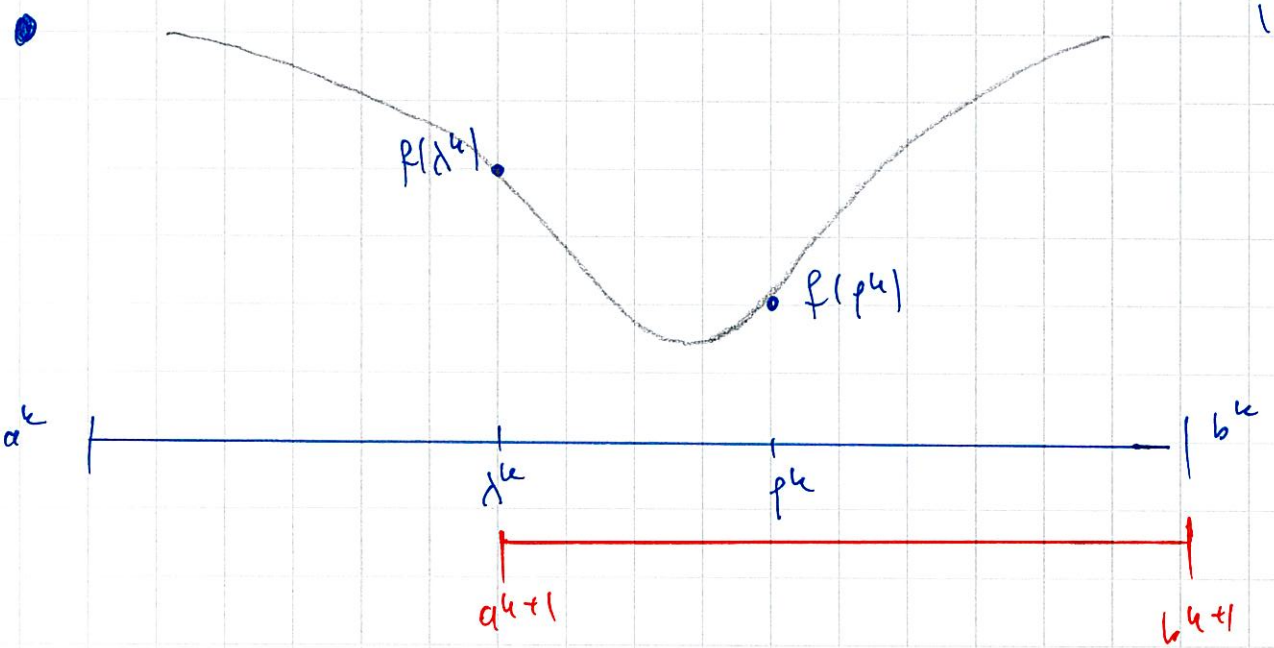


- Here's how we will reduce the interval of uncertainty iteratively:
- Say we are at iteration k and the current interval of uncertainty is $[a^k, b^k]$
- Take two points in the interval of uncertainty, a left point λ^k and a right point ρ^k.
- Compute $f(\lambda^k)$ and $f(\rho^k)$



If $f(\lambda^k) \leq f(\rho^k)$ then

$$a^{k+1} = a^k$$
$$b^{k+1} = \rho^k$$



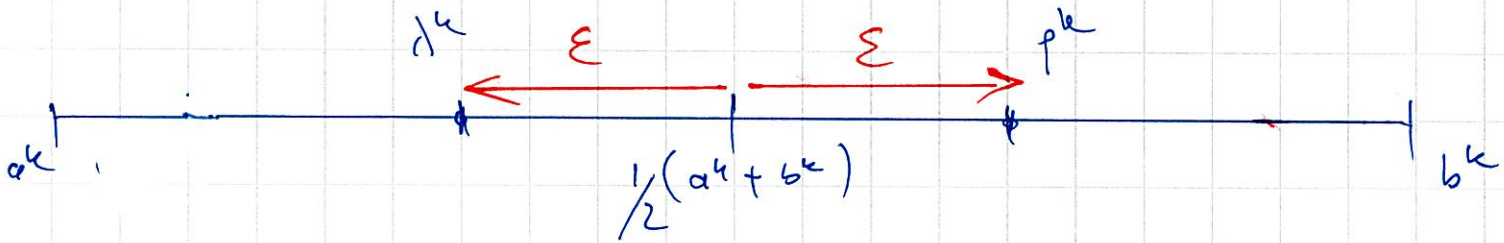
If $f(\lambda^k) > f(\rho^k)$ then

$$a^{k+1} = \lambda^k$$
$$b^{k+1} = b^k$$

Bisection Search For Single Dimensional Minimization

- Different ways of choosing λ^k and ρ^k gives us different algorithms.

- Here's one way of picking λ^k and ρ^k.
  Place λ^k and ρ^k symmetrically around the mid point of the interval of uncertainty



- If we choose ε very small, we need to compare $f(\lambda^k)$ and $f(\rho^k)$ but when λ^k and ρ^k are very close, we run into numerical difficulties when comparing $f(\lambda^k)$ with $f(\rho^k)$

- If we choose ε very large, then the interval of uncertainty will shrink slowly from one iteration to the next.